

LLMs Augmented Generative Design

Team 3

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Team Member : Wing, Alexis, Hendry

AI Hackathon 2025



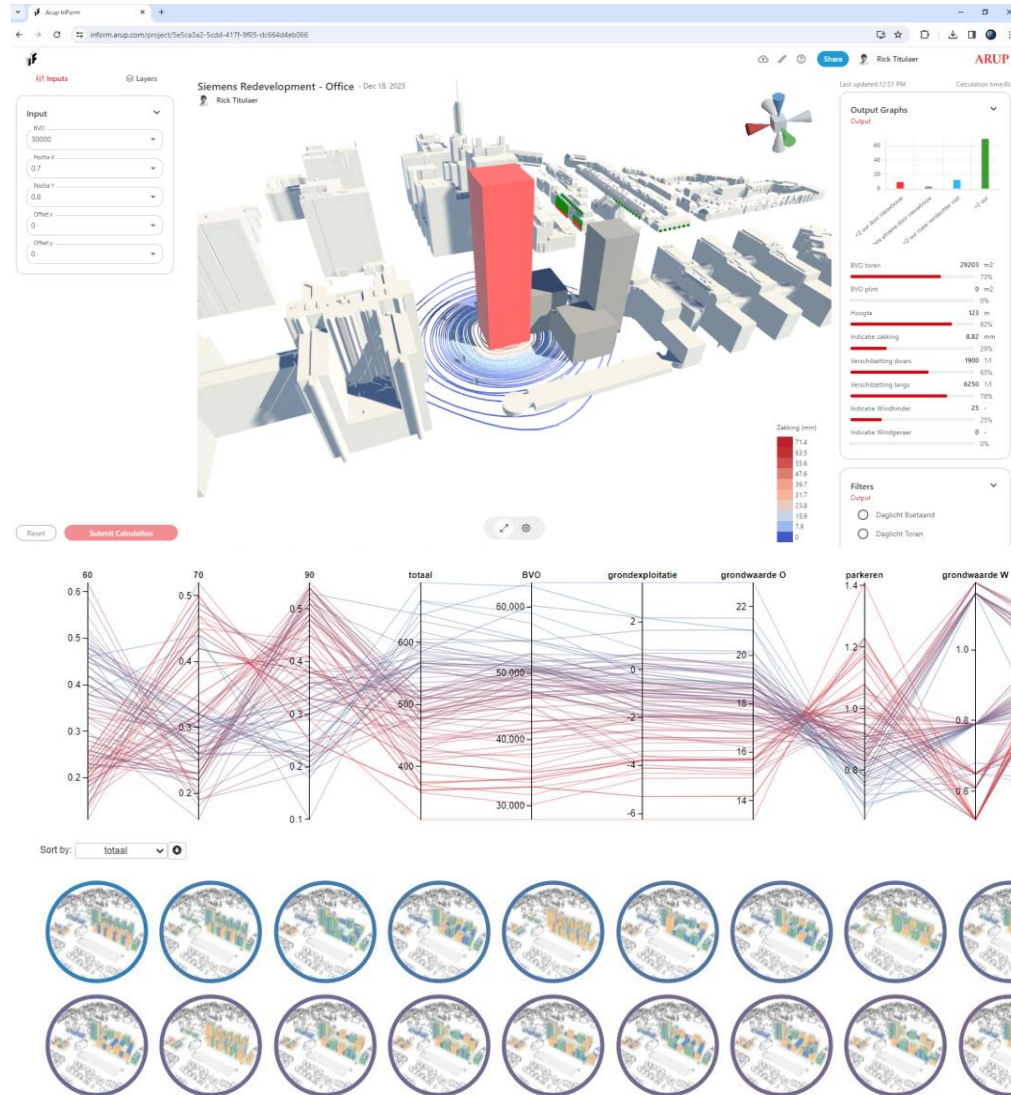
What's Computational Design



Generate and evaluate design options with parametric modeling

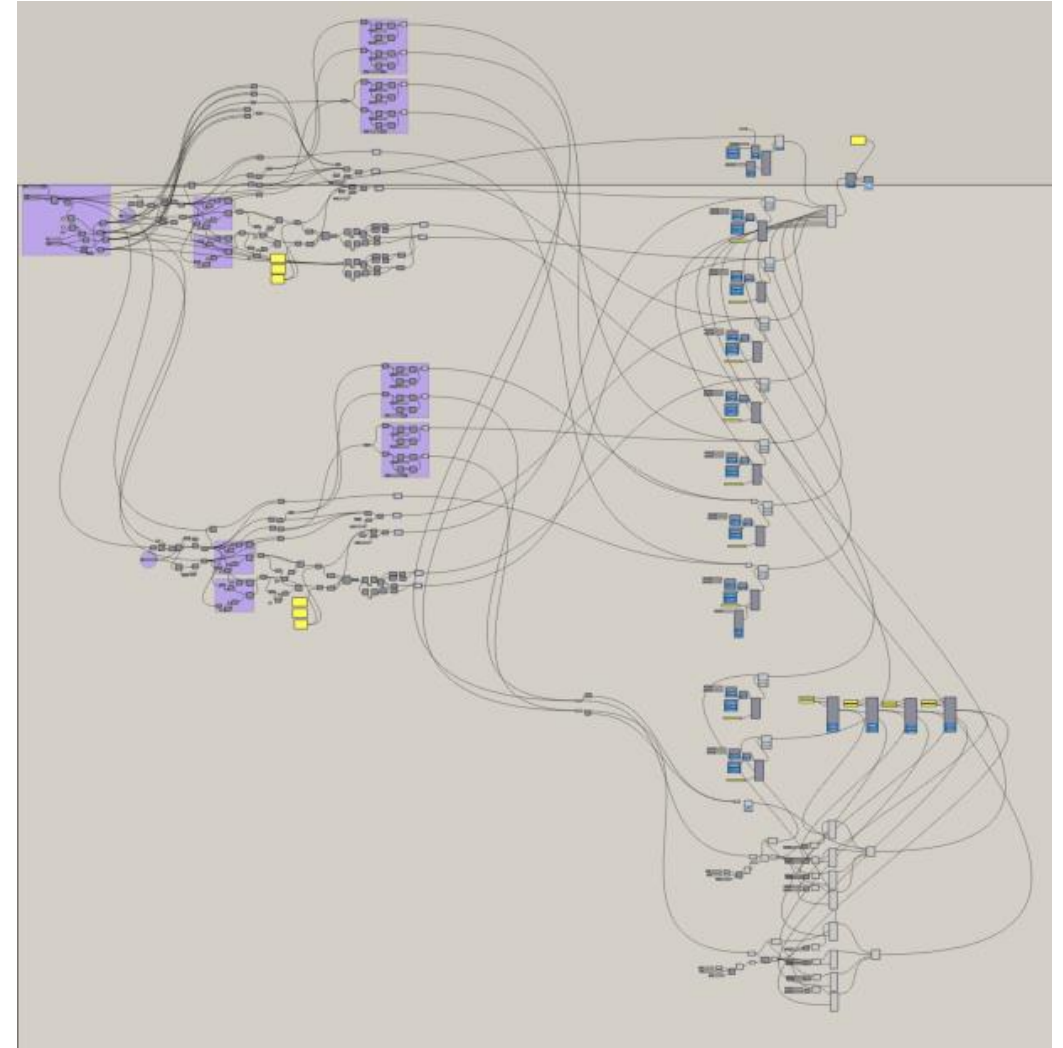


Explore the design space for optimized and balanced solution



Pain Point

- Difficult to transfer non-structured data into parametric workflow (e.g. sketches)
- Time consuming to build up the parametric workflow (normally take days) and difficult for others to learn how to use it.
- Massive domain knowledge required for the different plug-ins and functions

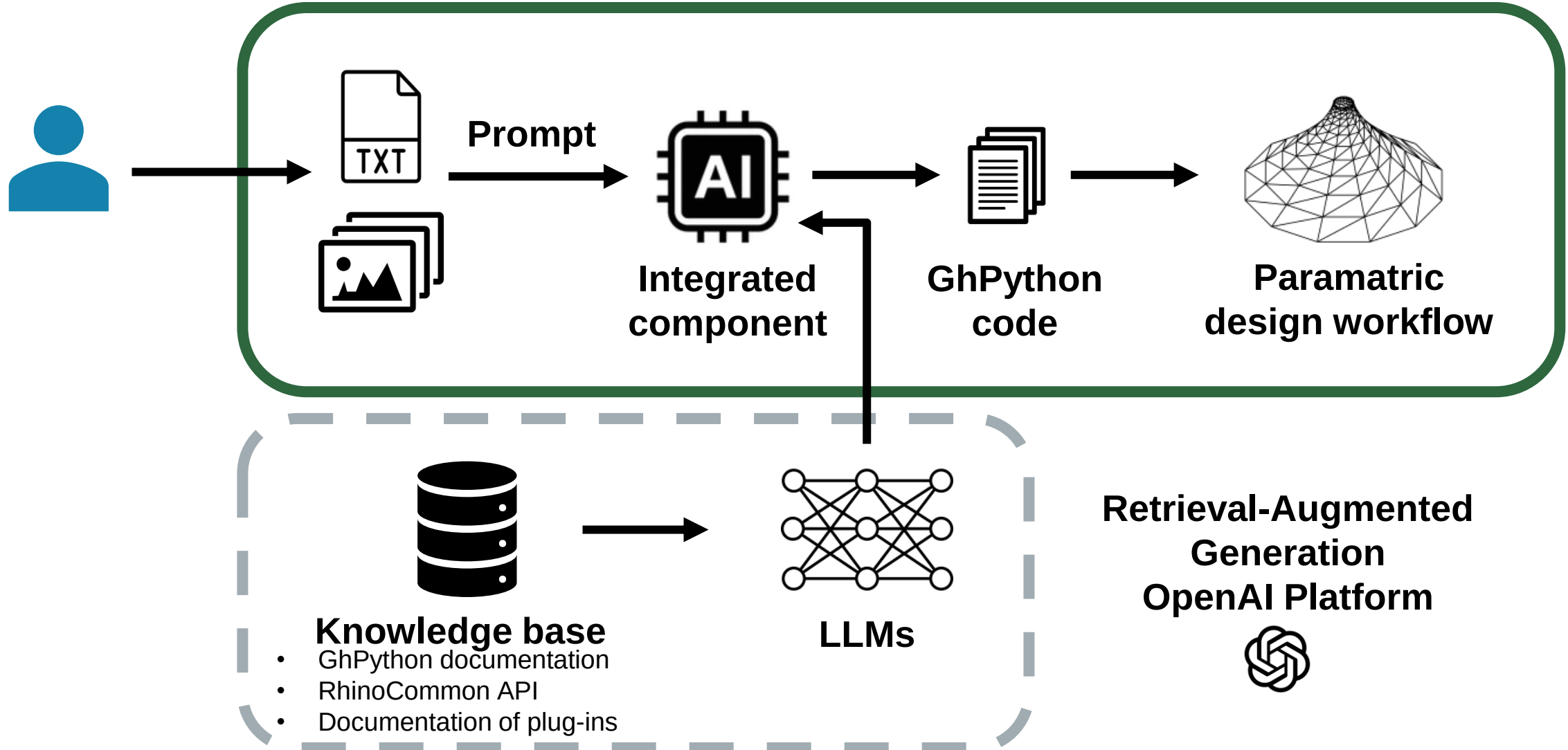


How can we accelerate the set-up process leveraging LLM?

- Can much geometrical information can LLM / VLM extract from images?
 - How could we deploy LLM agent within Grasshopper Environment to interact with existing parametric design scrips?
 - To what extend LLM can help us generate GhPython codes for complex functions?
 - Can we incorporate RAG to increase accuracy of the generated GhPython codes?
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- | | |
|---|---|
| ➤ Use LLMs to generate GHpython code to achieve logic that need multiple components | ➤ Leverage LLMs to recognize sketches and generate initial geometry |
|---|---|



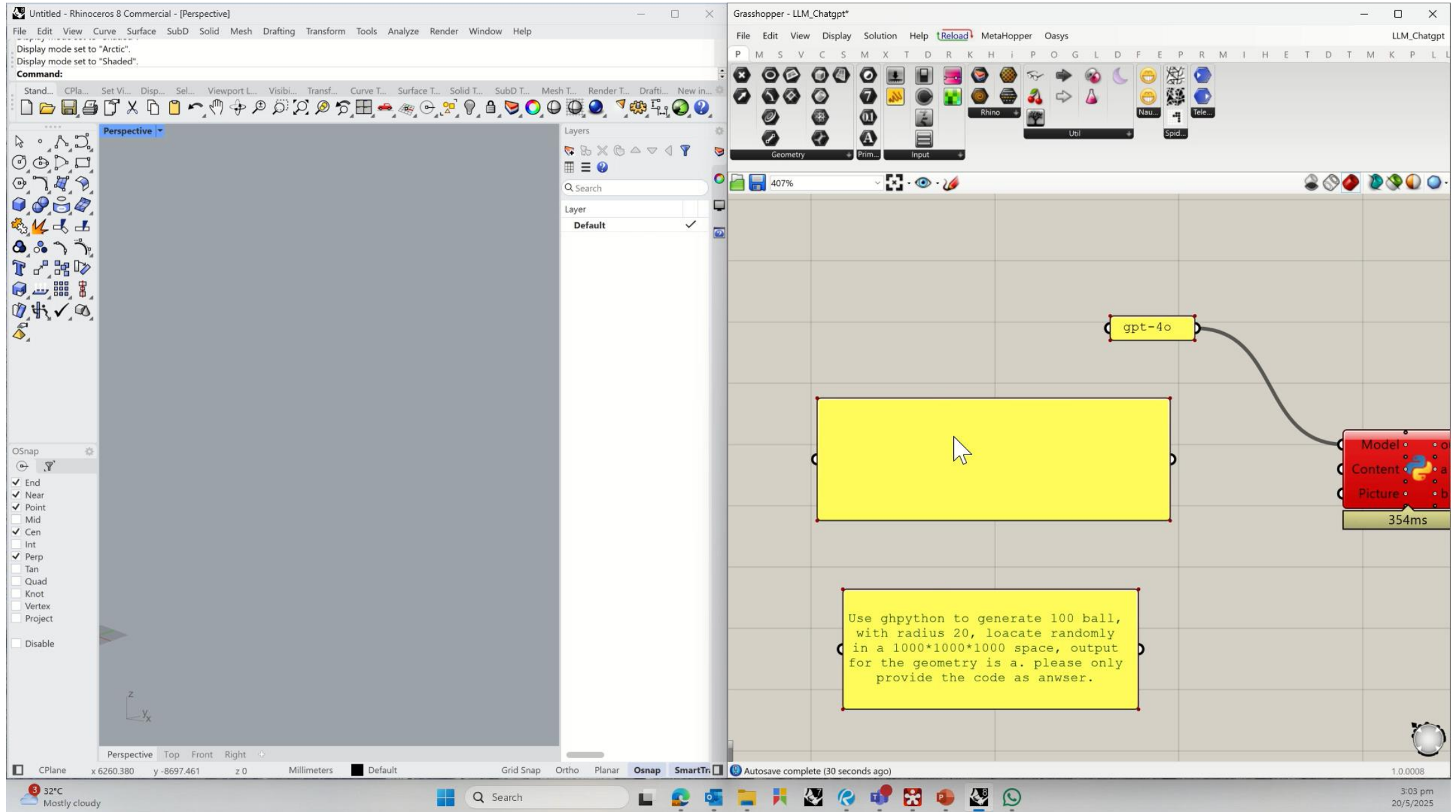
Rhino and Grasshopper
environment



LLMs Model Comparison

Model	Accuracy	Speed	Cost
GPT-4.1	4/5	4/5	\$\$\$\$\$
GPT-4o	5/5	4/5	\$\$\$\$\$
o4-mini (reasoning)	5/5	3/5	\$\$\$\$
GPT-4.1 mini	3/5	5/5	\$\$\$
QVQ-Max (reasoning)	4/5	1/5	\$
Qwen-VL	3/5	2/5	\$
Qwen-VL-Max	3/5	2/5	\$

Test & Measures



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名称 ↓

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- VO1_I_LeichtbauReview.pdf .pdf
- Snipaste_2025-05-20_20-19-53...
- Snipaste_2025-05-20_17-53-57...
- Snipaste_2025-05-18_16-40-01...
- Snipaste_2025-05-18_17-54-36...
- Snipaste_2025-04-24_20-06-11...
- Praktikumsbericht1.docx
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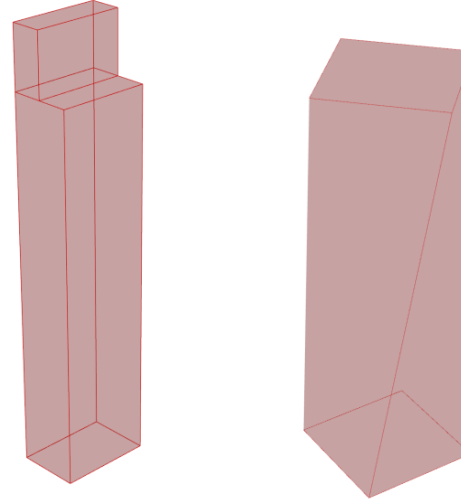
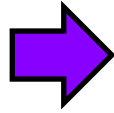
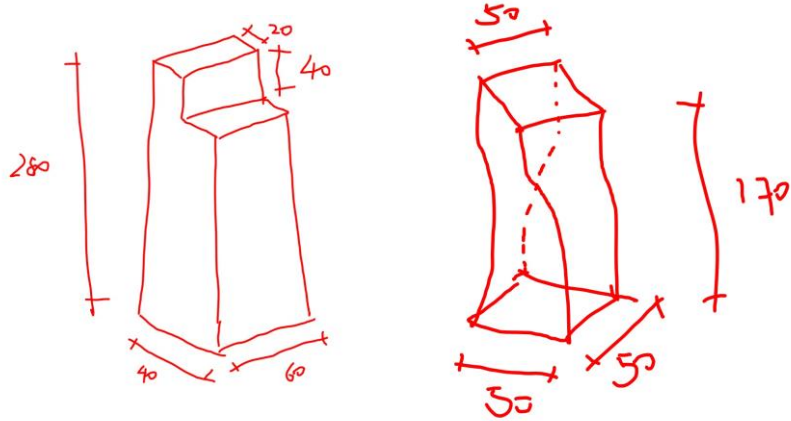
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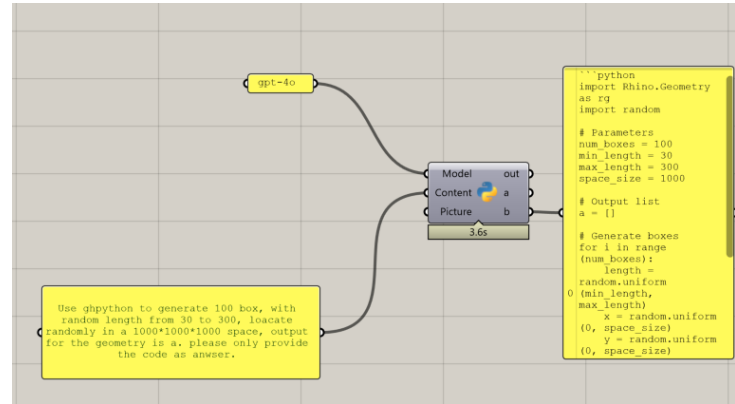
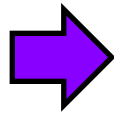
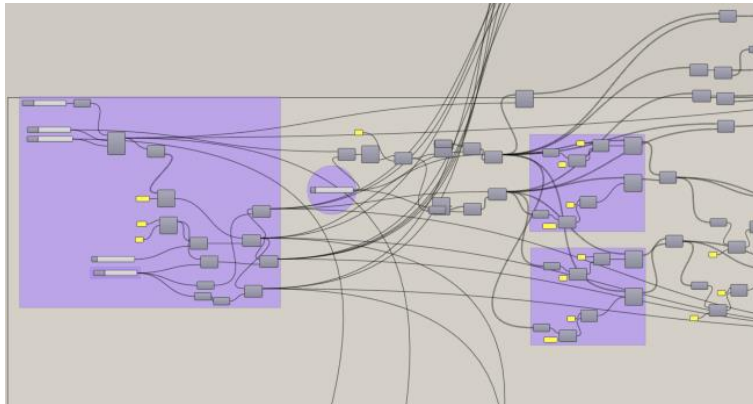
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修改日期 ↓ 文件大小

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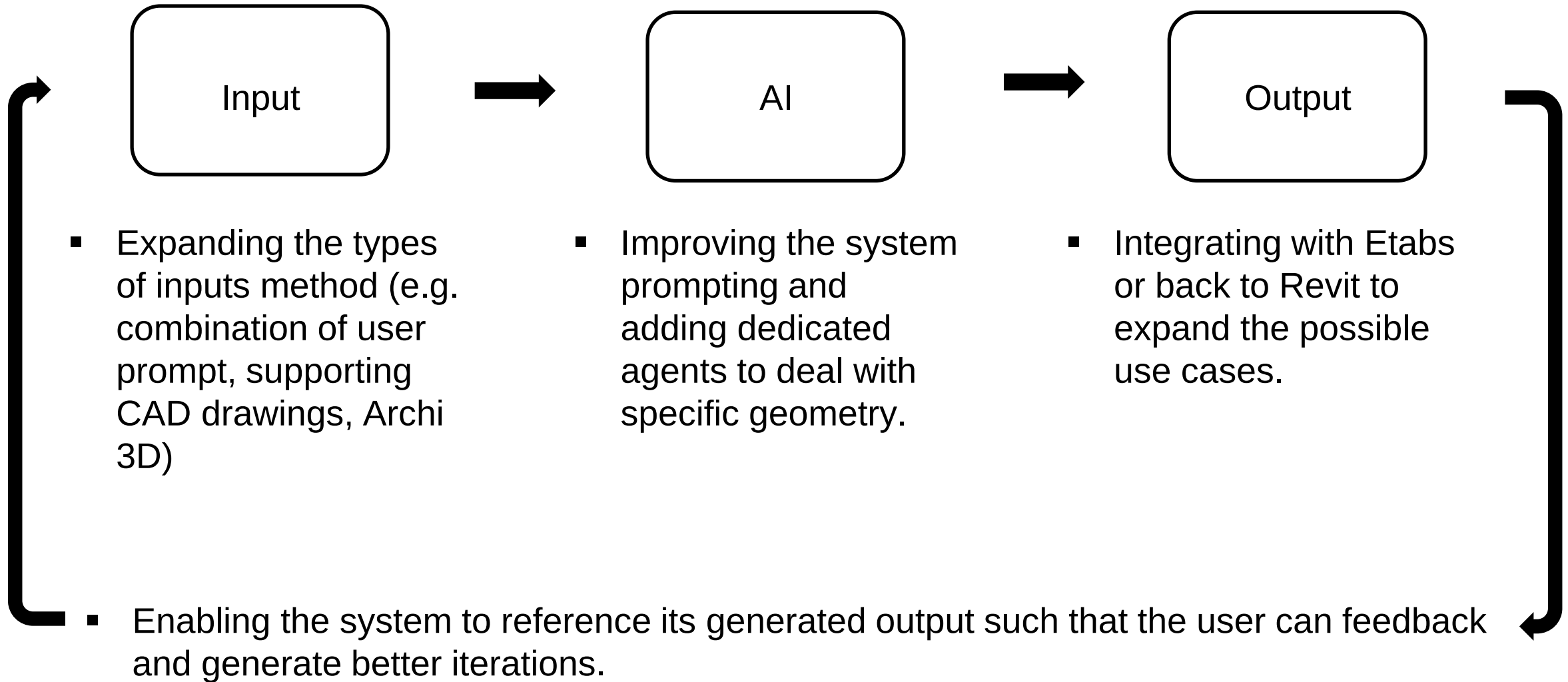
Sketches can be recognized and transferred into 3D geometries



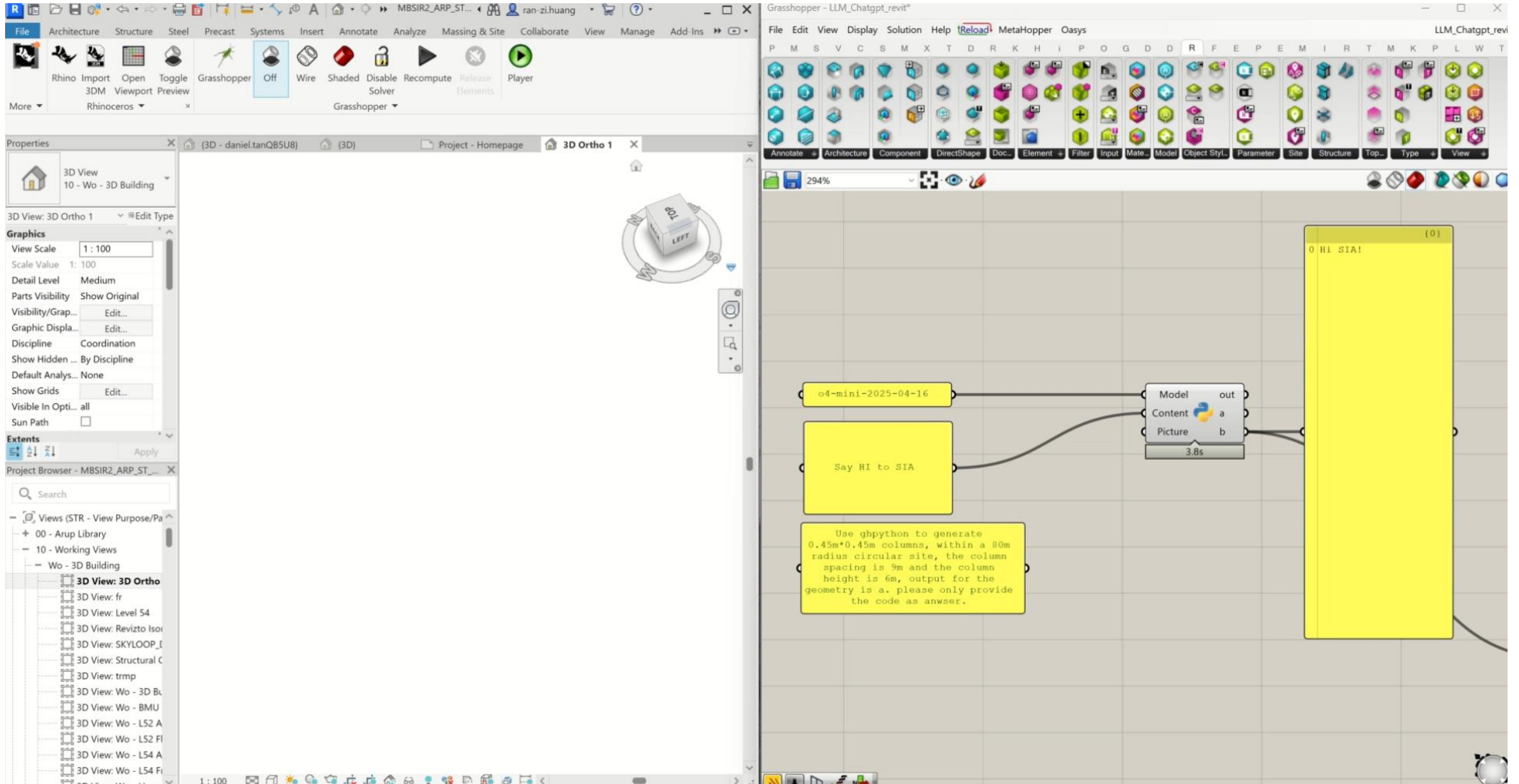
Complex parametric model set up process can be replaced by python code generated by LLMs

➤ Accelerate the setup time of computational design workflow by ~50%*

*Depends on the complexity of the workflow



What Next



- Image and drawing are fundamentally different "languages," and current LLMs still struggle with fully understanding architectural drawings and their spatial relationships.
- It is still unclear what GenAI is capable of, after all it is just trying to regurgitate what they are given to train and how. (no one knows)
- That said, there's growing interest in combining BIM data, floor plans, and text notes into unified reasoning systems — something that could drastically improve spatial understanding in AI.
- What's missing is an accessible common platform / technique for domain experts to test and experiment (such as Model Content Protocol (MCP)).

To be added: Link of the component (with API key removed)

Thank you